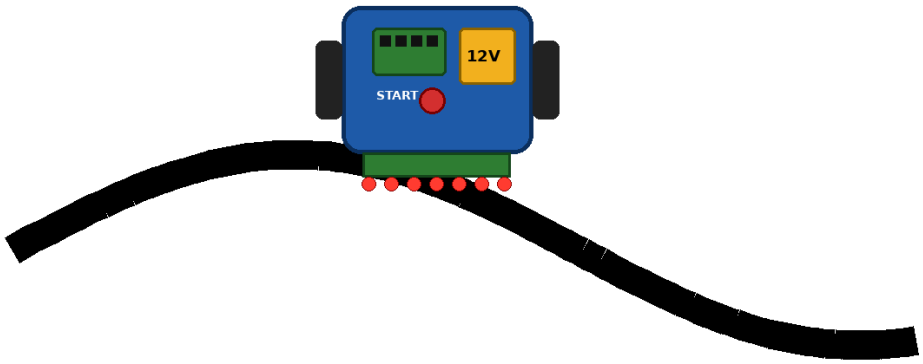
LINE FOLLOWER ROBOT COMPETITION

Official Rule Book

Line Follower Robot on Track

Black line | White surface | IR sensor array



1. Introduction

The Line Follower Robot Competition challenges autonomous robots to follow a black line on a white surface in the shortest time. The competition will test accuracy, speed, and robustness.

2. Team Eligibility

2.1 A team can consist of 2-4 members.

2.2 All members must be registered students.

2.3 One robot per team is allowed. The same robot cannot be used by two teams.

2.4 Team members can only be part of one team.

3. Robot Specifications

3.1 The robot must be fully autonomous. No remote control, Bluetooth, or manual control is allowed.

3.2 Robot dimensions: maximum 1.5 ft by 1.5 ft, with a maximum height of 0.5 ft.

3.3 Power supply must be onboard. Maximum 12V battery allowed.

3.4 Robot must have an onboard start/stop button.

4. Track Specifications

4.1 Track surface: white flex / matte white background with a black line.

4.2 Line width: 2.5 cm (1 inch) ± 0.5 cm.

4.3 The track will contain straight lines, curves, 90-degree turns, U-turns, and cross intersections. There will be NO broken lines or acute angles less than 90 degrees.

4.4 Start line and Finish line will be clearly marked.

4.5 Ambient lighting may vary. Teams must calibrate sensors accordingly.

5. Competition Format (3 Rounds)

Round 1: Basic Testing Round (Qualifying Round)

5.1 Purpose: to check if the robot is functional and can follow a line.

5.2 Each team will get a simple straight + curve test track.

5.3 Each team gets 3 attempts and must complete the track in at least 1 attempt.

5.4 Time limit: 3 minutes per attempt.

5.5 Teams who complete the track will qualify for Round 2. There is no time ranking in this round.

Round 2: Speed Round (Time Trial)

5.6 This is the main time-trial round on the full main track.

5.7 Each team gets 2 official runs. The best time of the 2 runs will be considered.

5.8 Time starts when the robot crosses the START line and stops when it crosses the FINISH line.

5.9 Maximum time limit per run: 5 minutes. If the robot does not finish in 5 minutes, the run is DNF (Did Not Finish).

5.10 The top 8 teams with the shortest times will qualify for Round 3.

Round 3: Final Showdown (Top Teams Battle)

5.11 All top 8 teams will race SIMULTANEOUSLY on the same track.

5.12 Robots will start with a gap of 50 cm from each other at the start line.

5.13 The first robot to complete 1 full lap wins. 2nd and 3rd positions will be decided similarly.

5.14 If a robot goes off-track or collides and stops, it will be eliminated from that race.

5.15 Deliberate pushing or blocking of other robots is strictly prohibited.

6. Rules & Penalties

6.1 Only 1 team member can be near the track during the run for start/stop.

6.2 No touching the robot after it starts unless permitted by the referee.

6.3 If the robot goes off the line for more than 3 seconds, that attempt will be considered failed.

6.4 Teams are allowed to re-calibrate for 2 minutes before their run.

6.5 The referee's decision is final.

7. Scoring

7.1 Round 1: Pass / Fail only.

7.2 Round 2: Ranking based on shortest completion time.

7.3 Round 3: Ranking based on finishing order.

7.4 In case of a tie, a re-run will be conducted.

8. Prizes & Certificates

8.1 The winner will receive a trophy, cash prize, and certificate (cash prize will be declared later).

8.2 All participants will receive a participation certificate.

9. General Conduct

9.1 Any misbehavior or argument with organizers/referees will lead to disqualification.

9.2 Teams must report 15 minutes before their slot.